

ICLR 2026 · code & data released (MIT)

DataProphet

Can you predict — *before any training* — how much a given supervision dataset will help a given target benchmark?

- training-free
- data influence
- multimodal
- Kendall's τ 0.860

The paper

- **Demystifying Supervision Data Generalization in Multimodal LMs** — ICLR 2026
- Xuan Qi (UPenn / Tsinghua), Luxi He (Princeton), Dan Roth (UPenn), Xingyu Fu (UPenn / Princeton)
- Code **MIT**: `github.com/DataProphet26/dataprophet` · data: HF THUQiXuan/DataProphet
- Project page: `dataprophet26.github.io`
- Completed during Xuan Qi's summer internship at UPenn

A measurement paper first and a method paper second — the empirical analysis is the stronger half.

The question, stated precisely

Given a training dataset D_i , can we predict its influence on a target benchmark T_j **before any training takes place?**

- Influence is measured as **relative performance change**: $\Delta_{s \rightarrow t} = (A_t^s - A_t) / A_t$
- i.e. how much target t improves after fine-tuning on source s
- The conventional approach is to trust **intuitive task similarity** — text-rich vs vision-centric

The paper's job is to test whether that convention survives contact with data.

The setup: a complete 14×14 matrix

Design

- Base model **InternVL3-2B**
- **20K** train / **1K** test per source
- **Fixed compute:** same data volume, model, optimizer, scheduler, epochs, batch size
- Full-parameter SFT, 1 epoch, lr $1e-5$ (LLaMA-Factory)

14 datasets, 7 task families (2 each)

- **OCR:** OCR-VQA, ScreenQA
- **Chart:** ChartQA, Chart2Text
- **Document:** DocVQA, FinanceQA
- **General VQA:** A-OKVQA, VC-GVQA
- **Spatial:** Open-Spatial, CLEVR-Relation
- **Counting:** CLEVR-Counting, TallyQA
- **Map:** GeomVerse, MapQA

Every claim in the paper rests on this one matrix — measured, not assumed.

Why this question is worth asking

- Data is one of the biggest levers on MLLM quality, and we have far more supervision datasets than budget.
- Prior methods **improve target performance by removing** irrelevant, redundant or low-quality data — most still need *some* training to decide.
- Domain mixtures are usually set by **intuition plus a proxy sweep**, and proxy sweeps are expensive.
- If influence could be **scored without training**, mixture design becomes a cheap, auditable decision.

That is the prize: turn "which data should we train on?" from an experiment into a calculation.

Finding 1 — influence is *asymmetric*

Open-Spatial → DocVQA

15.92%

DocVQA → Open-Spatial

26.14%

- $\Delta_{s \rightarrow t} \neq \Delta_{t \rightarrow s}$ — influence is not a symmetric similarity score.
- Diagonal gains are still largest (same-distribution data remains most influential).

Implication: influence is a *directed graph*, not a scalar property of a dataset. Any single "how good is this dataset" number discards this by construction.

Finding 2 — same task family $\neq$ most help

After training on **OCR-VQA**:

ScreenQA — also OCR

+17.88%

GeomVerse — map understanding

+21.74%

Training on OCR helps a *map* task more than it helps another *OCR* task. The paper's phrasing: influence is decided by the **individual dataset**, not the task category.

Finding 3 — text-rich can help vision-centric most

The intuition: OCR and charts both need text and numbers extracted from images, so OCR data should help chart tasks most.

After training on **ScreenQA** (an OCR dataset):

Text-rich targets

- ChartQA **+5.61%**
- Chart2Text **+6.13%**

Vision-centric targets

- Open-Spatial **+12.86%**
- CLEVR-Relation **+7.96%**

The result is the **reverse** of the intuition. Surface similarity does not predict transfer.

Three findings, one conclusion

1. **Influence is asymmetric** — $\Delta_{s \rightarrow t} \neq \Delta_{t \rightarrow s}$.
2. **Same task family $\neq$ strongest mutual help** — it is dataset-specific, not category-specific.
3. **Text-rich can help vision-centric more than text-rich** — the intuition points the wrong way.

Intuition about task similarity is an unreliable guide to transfer. So you need a predictor you can actually *compute*.

DATAPROPHET, in one equation

$$M(s \rightarrow t) = QSim \cdot ASim \cdot ISim \cdot PPL(s) \cdot (Sil + H) / PPL(t)$$

- **QSim / ASim / ISim** — question, answer and image similarity between source and target
- **PPL(s) / PPL(t)** — multimodal perplexity of the source, divided by the target's
- **Sil + H** — source question diversity (silhouette + cluster-balance entropy)
- All three components are computed with the **frozen base model** — no training anywhere

Component 1 — multimodal perplexity

$$\text{PPL}(D) = \exp(- E_{(I,Q,A) \sim D} E_t \log p_{\theta}(a_t \mid I, \tau_Q(Q), a_{<t}))$$

- **Hypothesis:** intrinsically *harder* source data provides greater value for enhancing capability.
- Precedent in LLM work: perplexity as a data-quality estimator for corpus pruning.
- **Alone it is not enough:** perplexity-only gives $\tau_{\text{Tgt}} = \mathbf{0.274}$.

It is, however, the single most important factor once combined — see the ablation.

Component 2 — cross-dataset similarity

- If source data resembles the target in the model's **embedding space**, it should be closer to "learning the target distribution".
- Use the base MLLM's **frozen encoders**; encode **questions, answers and images separately** with templates; ℓ_2 -normalize; take expected cosine similarity $\rightarrow$ **QSim, ASim, ISim**.

A negative result worth recording: encoding question and answer *jointly* as QASim was **6% worse** in τ than keeping them separate.

Still not enough on its own: PPL $\times$ similarity gives $\tau_{\text{Tgt}} = \mathbf{0.658}$.

Component 3 — source question diversity

- **Hypothesis:** a source whose questions cover more ground encourages more generalizable skills.
- Embed each source question, **K-means into K = 10** clusters, then measure two things:

Silhouette — separation

$$\text{Sil} = E[(b(u) - a(u)) / \max\{a(u), b(u)\}]$$

$a(u)$ = mean intra-cluster distance; $b(u)$ = nearest other cluster.

Entropy — balance

$$H = -(\log K)^{-1} \sum \pi_k \log \pi_k$$

π_k = empirical cluster proportion.

Diversity = **Sil + H**: larger means broader coverage with better-separated, better-balanced clusters.

Why a *product*, and how it is scored

Successful transfer requires **simultaneous** alignment in text, vision, difficulty relative to the base model, and question coverage. The product form therefore **down-weights the score when any single factor is weak**.

- A weighted **sum** would let a strong factor mask a weak one. A product does not.
- **Two-way evaluation protocol** (because influence is asymmetric):
 - τ_{Tgt} — fix a target, rank the 14 sources, compare with reality
 - τ_{Src} — fix a source, rank the 14 targets, compare with reality
 - final $\tau = (\tau_{\text{Tgt}} + \tau_{\text{Src}}) / 2$

Does it predict the rankings?

τ_{Tgt} average

0.863

range 0.736 – 0.933

τ_{Src} average

0.857

range 0.822 – 0.911

Headline

86.0%

Kendall's τ , both directions

- Consistently high across all 7 task families — not driven by one easy target.
- The metric captures the **asymmetry**, which a symmetric similarity score could not.

Ablation: what actually carries the signal

- **Full DATAPROPHET** — τ **0.860**
- w/o Answer Similarity — 0.810
- w/o Question Similarity — 0.778
- w/o Diversity (Silhouette & Entropy) — 0.659
- w/o Image Similarity — 0.625
- **w/o Perplexity** — **0.487** (roughly halves the correlation)

Clear ordering: **perplexity** > **image similarity** $\approx$ **diversity** > **text similarity**. Textual similarity contributes least — consistent with Finding 3.

The heuristics that did *not* work

All of these were tried and none improved prediction:

- Question difficulty
- Model's familiarity with the image

- Model's familiarity with the question
- Answer length — adding it **decreased τ by 0.15**

Most "sensible-looking" data properties do not predict transfer. Publishing this list is more useful to successors than publishing only what worked.

From prediction to selection

Fixed budget: $N = 14 \times 20K = 280K$ samples. Reweight each source by its DATAPROPHET score.

Real data

- Uniform — 67.6
- **ICONS** (training-based SoTA) — 69.6 (+2.0)
- Oracle (observed gains) — 70.8 (+3.2)
- **D.P. — 71.0 (+3.4)**

Synthetic data (~1.4M pool)

- Uniform — 55.1
- ICONS — 60.8 (+5.7)
- **D.P. — 62.0 (+6.9)**

D.P. beats the training-based selector in both settings — and the paper reports it **edging the Oracle by 0.2%**, which is the claim to scrutinise.

It also works for RL, and ranks the generators

- **GRPO** via **verl** on Qwen2.5VL-3B-Instruct, prompt budget 300×14:
 - Real RL data: no-RL 55.7 → Equal 58.3 → **D.P. 59.5**
 - Synthetic RL data: baseline 56.4 → Random 56.4 → **D.P. 57.7**
- So the SFT conclusion **transfers to post-training** — a genuine broadening.

A neat by-product: of the selected synthetic items, **38% came from GPT-5 and 62% from Gemini 2.5 Pro** — the selector implicitly ranks Gemini's data higher.

What to take

- **Treat influence as a directed quantity.** $\Delta_{s \rightarrow t} \neq \Delta_{t \rightarrow s}$ is measured, so a single quality score per dataset is structurally insufficient. *This is the big one.*
- **"Looks similar" is not evidence.** If your mixture weights come from task-similarity intuition, this paper says that heuristic is unreliable.
- **Perplexity is the best single cheap signal** — and it points the same way as the dedup finding that high-frequency repetition means low information density.
- **Multiply when conditions must hold jointly.** Sums hide weak factors.
- **Publish the failed heuristics.** Four of them here, all plausible, none predictive.

What to discount

- **"Beats the Oracle by 0.2%" is not a real result.** No seeds, no variance, no confidence intervals — 0.2% on a 14-benchmark macro average is indistinguishable from noise. Worse, the Oracle is built from **20K-per-source** observations and extrapolated to a 280K mixture, exactly as D.P. is, so it is not a strict upper bound.
- **Everything rests on one 2B model** (InternVL3-2B). The matrix, the ablations, the gains — no second family, no larger model.
- **"Training-free" is not "model-free."** PPL uses the base model and the similarities use its frozen encoders, so **scores do not transfer across models** — change the base and recompute everything.
- **2 datasets per task family** is thin support for a general claim about task *categories*.
- **Small scale:** 20K samples, 1 epoch. Real instruction tuning is 100K–millions, multi-epoch.
- **The Gemini-vs-GPT-5 comparison is confounded** — the two generators saw *different* image halves.

Takeaways for our own work

- **Model data value as a matrix, not a vector.** If you must collapse it, be explicit about which direction you are collapsing.
- **Budget for a cheap scorer.** Perplexity over candidates plus one embedding pass is far cheaper than a proxy sweep — but it is not free at millions of candidates.
- **Write down the base-model dependency.** Any embedding- or perplexity-based score is bound to the model that produced it; that has to be stated before anyone plans to reuse it.
- **Use a selector's preferences to rank sources.** "62% of picked items came from Gemini" is a lightweight, useful way to compare data generators.
- **Reproducibility is achievable** — MIT code, public dataset, project page. Worth matching.

Full analysis: `my-wiki/05-References/DataProphet` 论文总结（中文）.md and the English counterpart.